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MULTISPECTRAL SATELLITE IMAGE DEHAZING

● **MINI PROJECT -1B**

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○ ABSTRACT

In this Project and Paper, we present an end-to-end network, called Cycle-Dehaze, for single image dehazing problem, which does not require pairs of hazy and corresponding ground truth images for training. That is, we train the network by feeding clean and hazy images in an unpaired manner. Moreover, the proposed approach does not rely on estimation of the atmospheric scattering model parameters. Our method enhances CycleGAN formulation by combining cycle-consistency and perceptual losses in order to improve the quality of textural information recovery and generate visually better haze-free images. Typically, deep learning models for dehazing take low resolution images as input and produce low resolution outputs. However, in the NTIRE 2018 challenge on single image dehazing, high resolution images were provided. Therefore, we apply bicubic down scaling. After obtaining low-resolution outputs from the network, we utilize the Laplacian pyramid to upscale the output images to the original resolution. We conduct experiments on NYU-Depth, I-HAZE, and O-HAZE with final custom dataset which carries variety of data which avoid overfitting and underfitting in the system datasets. Extensive experiments demonstrate that the proposed approach improves CycleGAN method both quantitatively and qualitatively.

LITERATURE SURVEY

2.1 Existing System

Image dehazing is a crucial task in satellite image processing, aimed at improving the visibility and quality of images affected by atmospheric haze. Several existing methods have been proposed to tackle this problem, ranging from traditional image processing techniques to deep learning-based approaches.

1. Traditional Image Processing Techniques

Early methods relied on traditional image enhancement techniques such as:

- Histogram Equalization (HE): Adjusts image contrast by redistributing pixel intensity. However, it often over-enhances certain regions.
- Dark Channel Prior (DCP): Proposed by He et al. (2009), this method estimates the haze transmission map by assuming that at least one color channel has low intensity in non-hazy regions. While effective, DCP struggles with bright sky regions and is computationally expensive.
- Retinex-Based Methods: These methods enhance local contrast but often introduce artifacts and noise in satellite images.

2. Deep Learning-Based Approaches

With advancements in deep learning, CNN and GAN-based models have been developed to address image dehazing more effectively.

- DehazeNet: A CNN-based approach that directly learns the transmission map from hazy images, significantly improving accuracy.
- Pix2Pix GAN: A supervised deep learning method that uses paired hazy and dehazed images to generate clear images with high fidelity.
- CycleGAN: Suitable for unpaired datasets, CycleGAN learns to transform hazy images into dehazed ones by preserving structural details.
- U-Net: A segmentation-based architecture that is commonly used for paired image restoration, making it effective in satellite image dehazing tasks.

3. Limitations of Existing Systems

Despite advancements, existing methods face several challenges:

- Loss of Detail: Some traditional methods fail to retain fine details in dehazed images.
- Computational Complexity: Deep learning models require high computational power, making real-time processing difficult.
- Dependency on Paired Data: Many supervised learning approaches require paired hazy and dehazed images, which are difficult to obtain for real-world satellite images.

PROBLEM STATEMENT

Satellite images play a crucial role in various applications such as environmental monitoring, disaster management, and urban planning. However, atmospheric conditions, particularly haze, significantly degrade the quality of these images, reducing visibility and affecting the accuracy of downstream tasks like object detection and land classification. Traditional image dehazing techniques, including histogram equalization and dark channel prior (DCP), often fail to generalize across different atmospheric conditions and may introduce artifacts or excessive smoothing. Deep learning-based approaches, such as Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs), have shown promise in improving dehazing performance. However, existing methods face challenges such as:

- Dependency on paired hazy and dehazed images for training, which are difficult to obtain in real-world satellite datasets.
- Loss of fine details and texture information in high-resolution multispectral images.
- High computational cost, making real-time or large-scale processing infeasible.

This project aims to develop an efficient deep learning-based satellite image dehazing model that leverages Pix2Pix GAN for paired image datasets and CycleGAN for unpaired datasets to enhance image quality while preserving fine details. The model will process hazy multispectral satellite images and generate clear, high-fidelity dehazed images suitable for real-world applications. Additionally, an integrated LLM-based alert system will provide insights based on dehazing results, aiding decision-making in critical applications such as weather forecasting, environmental analysis, and urban planning.

METHODOLOGY

The proposed Multispectral Satellite Image Dehazing methodology is based on deep learning techniques, utilizing Pix2Pix GAN for paired images and CycleGAN for unpaired images. The approach involves multiple stages, from data preprocessing to model deployment, ensuring efficient and high-quality dehazing.

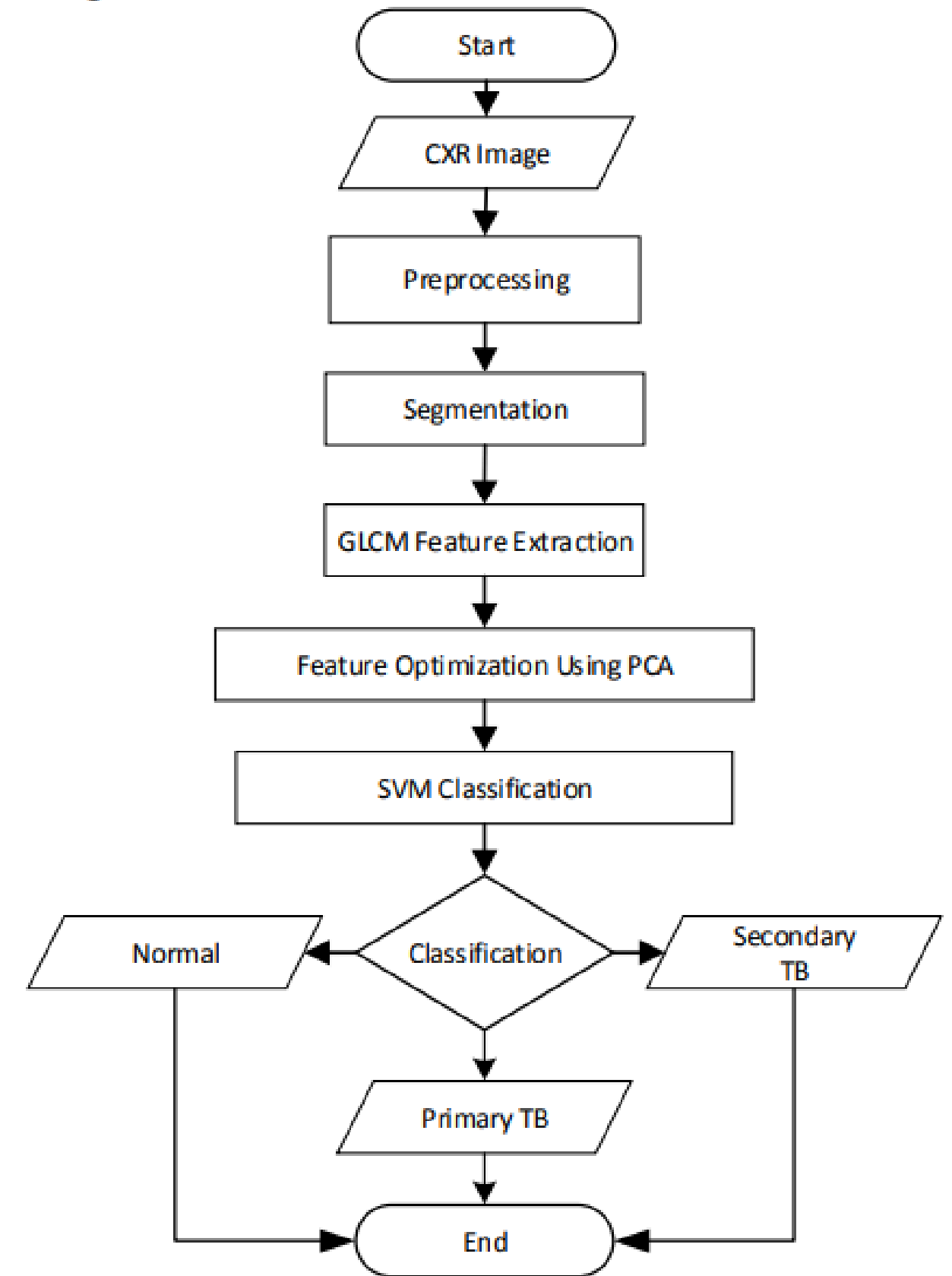
1. Data Collection & Preprocessing

1.1 Dataset Preparation

- The dataset consists of hazy and dehazed multispectral satellite images categorized into:
 - Paired Data: Hazy images with corresponding ground truth dehazed images (for Pix2Pix GAN).
 - Unpaired Data: Hazy and dehazed images without exact one-to-one correspondence (for CycleGAN).

1.2 Data Augmentation

- Techniques such as rotation, flipping, brightness adjustment, and noise addition are applied to increase dataset diversity and improve model generalization.
- Images are resized and normalized to meet input requirements for GAN architectures.



METHODOLOGY

2. Model Selection & Development

2.1 Pix2Pix GAN for Paired Data

- A Conditional GAN (Pix2Pix) is trained on paired images to learn a direct mapping from hazy to dehazed images.
- Generator: Uses a U-Net-based architecture to generate dehazed images.
- Discriminator: A PatchGAN classifier that ensures realistic output images.
- Loss Function: Adversarial Loss + L1 Loss to minimize structural inconsistencies.

2.2 CycleGAN for Unpaired Data

- A Cycle-Consistent GAN (CycleGAN) is used to translate hazy images into dehazed ones without requiring paired training samples.
- Forward Cycle Consistency ensures that an image translated to the target domain can be reconstructed back.
- Backward Cycle Consistency ensures that the transformation preserves structural integrity.
- Loss Function: Cycle Consistency Loss + Adversarial Loss to balance translation and realism.

3. Model Training & Optimization

- Hardware Utilization: Training is performed on high-performance GPUs to optimize computational efficiency.
- Hyperparameter Tuning:
 - Learning rate: 0.0002 (GAN) with Adam optimizer
 - Batch size: 8–16
 - Training epochs: 100+ for stability
- Evaluation Metrics:
 - Peak Signal-to-Noise Ratio (PSNR)
 - Structural Similarity Index (SSIM)
 - Frechet Inception Distance (FID) for perceptual quality

METHODOLOGY

4. Integration of LLM-Based Alert System

- An offline Large Language Model (LLaMA) is integrated into the pipeline.
- The LLM analyzes dehazing results and provides alerts based on significant changes in image clarity, useful for applications like disaster management or environmental monitoring.

5. Model Deployment

- The trained model is deployed using Django as a web-based application.
- Users can upload hazy images, get real-time dehazing results, and download processed images.
- The system also generates automated reports based on dehazing insights.

Conclusion

This methodology ensures that the proposed model efficiently removes haze from multispectral satellite images while maintaining structural integrity and detail preservation. By integrating both Pix2Pix GAN (paired) and CycleGAN (unpaired) models, the system achieves robust and generalizable performance across different datasets. Additionally, the LLM-based alert system enhances decision-making capabilities for real-world applications.

REQUIRED TECHNOLOGY

1. Programming Languages

Python – For implementing deep learning models and backend development.

2. Deep Learning Frameworks

TensorFlow / Keras – For building and training GAN-based models (Pix2Pix, CycleGAN).

PyTorch – Alternative framework for implementing CycleGAN with better flexibility.

3. Computer Vision & Image Processing

1. OpenCV – For image preprocessing, augmentation, and visualization.

2. Pillow (PIL) – For image handling and manipulation.

4. Machine Learning Libraries

1. NumPy – For numerical computations.

2. Scikit-image – For additional image processing functionalities.

3. Matplotlib / Seaborn – For visualizing results.

5. Dataset Management

HDF5 / TFRecords – Efficient storage and handling of large satellite image datasets.

6. GAN Models & Architectures

1. Pix2Pix GAN – For supervised learning with paired images.

2. CycleGAN – For unsupervised learning with unpaired images.

3. U-Net – For generator architecture in Pix2Pix.

7. Model Training & Optimization

1. Jupyter Notebook – For prototyping and experimentation.

2. TensorBoard – For monitoring training progress and performance.

8. LLM Integration for Alerts

LLaMA (Offline LLM) – For analyzing dehazing results and generating alerts.

9. Backend & API Development

1. Django / Flask – To build the web application and API.

2. FastAPI – Alternative for high-performance API deployment.

10. Frontend Development

1. HTML, CSS, JavaScript – For UI development.

2. Bootstrap / Tailwind CSS – For responsive design.

11. Deployment & Cloud Services

1. Docker – For containerizing the application.

2. AWS / Google Cloud / Azure – For cloud-based hosting and model inference.

3. SQLite / PostgreSQL – Database for storing results and user uploads.

CONCLUSION

- The proposed **Multispectral Satellite Image Dehazing system** leverages advanced deep learning techniques, including **U-Net**, **Pix2Pix GAN** for paired images and **CycleGAN** for unpaired images, to effectively remove haze while preserving structural details. The integration of an **LLM-based alert system (LLaMA)** enhances decision-making by analyzing dehazing results for critical applications such as disaster management and environmental monitoring.
- With a **Django-based web interface**, the system ensures user-friendly access, allowing users to **upload hazy satellite images and obtain high-quality dehazed outputs**. The use of **optimized loss functions**, and **cloud deployment** further enhances efficiency and scalability.
- This research and development contribute to improving remote sensing applications by providing clearer, high-fidelity satellite imagery, making it a valuable tool for government agencies, meteorologists, and researchers. Future enhancements can focus on real-time processing, adaptive dehazing techniques, and multimodal image analysis to further improve performance.

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